Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

Reduction of A, B pillar door hinge bolt slippage at Scorpio Bodyshop

Mahindra and Mahindra Ltd , Nasik Plant, Scorpio Bodyshop.

	Plan	Actual
Start Date	15.05.2012	15.05.2012
Completion Date	30.07.2012	30.07.2012

Current Status:

Completed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardise
- Step 9: Summarise & Plan future actions
- Guidelines to user

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	**SHINDE SANDIP S.**	DEPUTY MANAGER	SCORPIO BODYSHOP	9850291452	SHINDE.SANDIP @ mahindra.com
2	**Mentor**	**SURYAVANSHI R G**	DGM & HOD SCORPIO BODYSHOP	SCORPIO BODYSHOP	9881378988	SURYAVANSHI. RAVINDRA@ mahindra.com
3	**Member 1**	**GANGURDE M.**	ASST. MANAGER	SCORPIO BODYSHOP	9922644399	GANGURDE. MANGESH1@ mahindra.com
4	**Member 2**	**BHAT VIVEK**	DEPUTY MANAGER	SCORPIO BODYSHOP	9922438403	BHAT.VIVEK@ mahindra.com
5	**Member 3**	**DEVESH KUMAR**	DEPUTY MANAGER	SQA,NASIK	8888814357	KUMAR.DEVESH @ mahindra.com
6	**Member 4**	**GONSALVES S.**	Plant Head	M/s Harshal Pressing, Chakan	9881158509	s.gonsalves@ waradegroup.com'

1. Select the Theme

Brief description of Problem

What: **'A' plr & 'B' plr door hinge bolt slippage issue**

Where: **FRT & RR DR fitment at closure fitment stage**

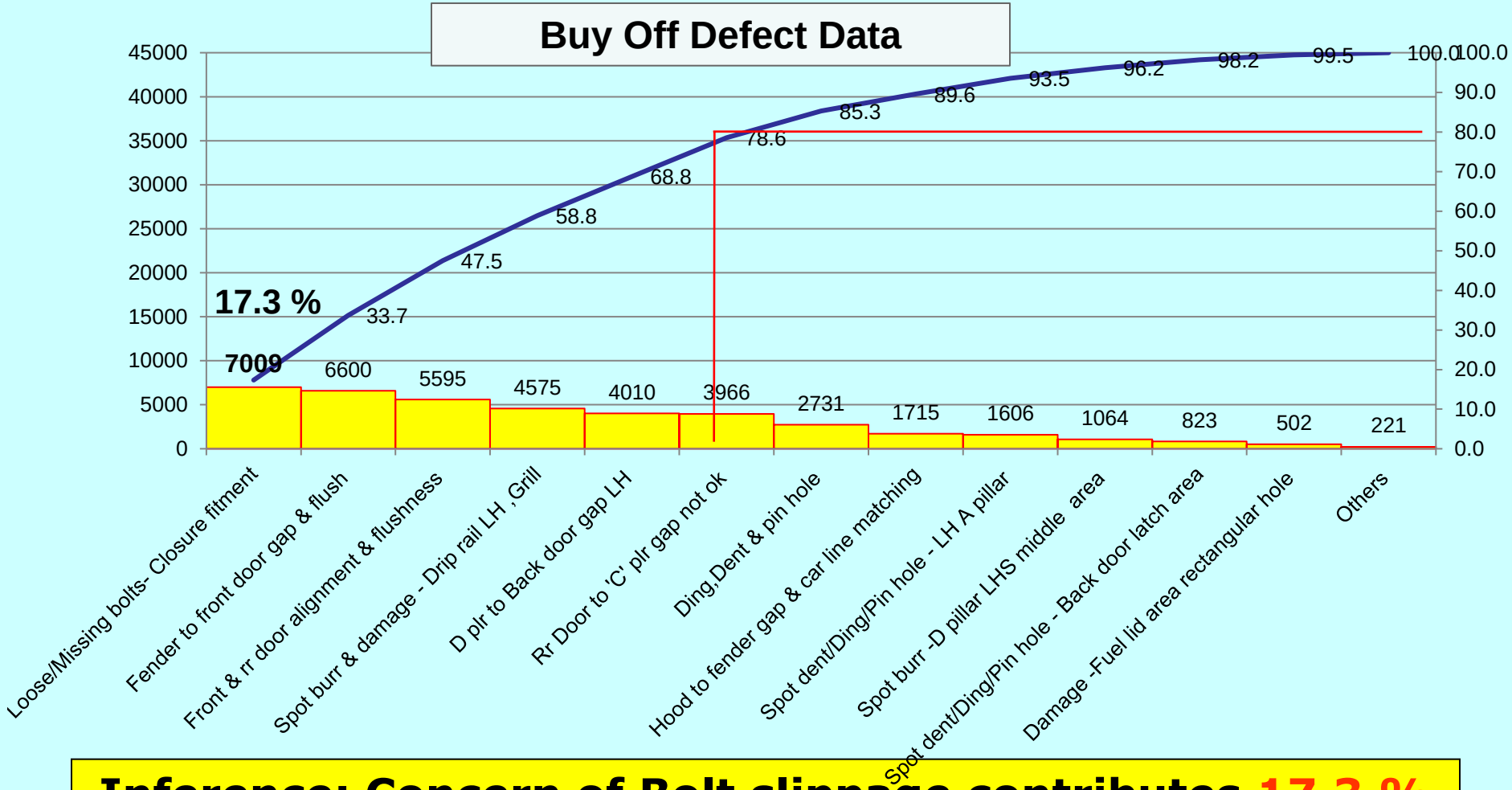
Who: **Concern reported by closure fitment stage & Buyoff operator, check man**

When: **Daily (1st and 2nd shift)**

Why: **high rework incidences & rework cost**

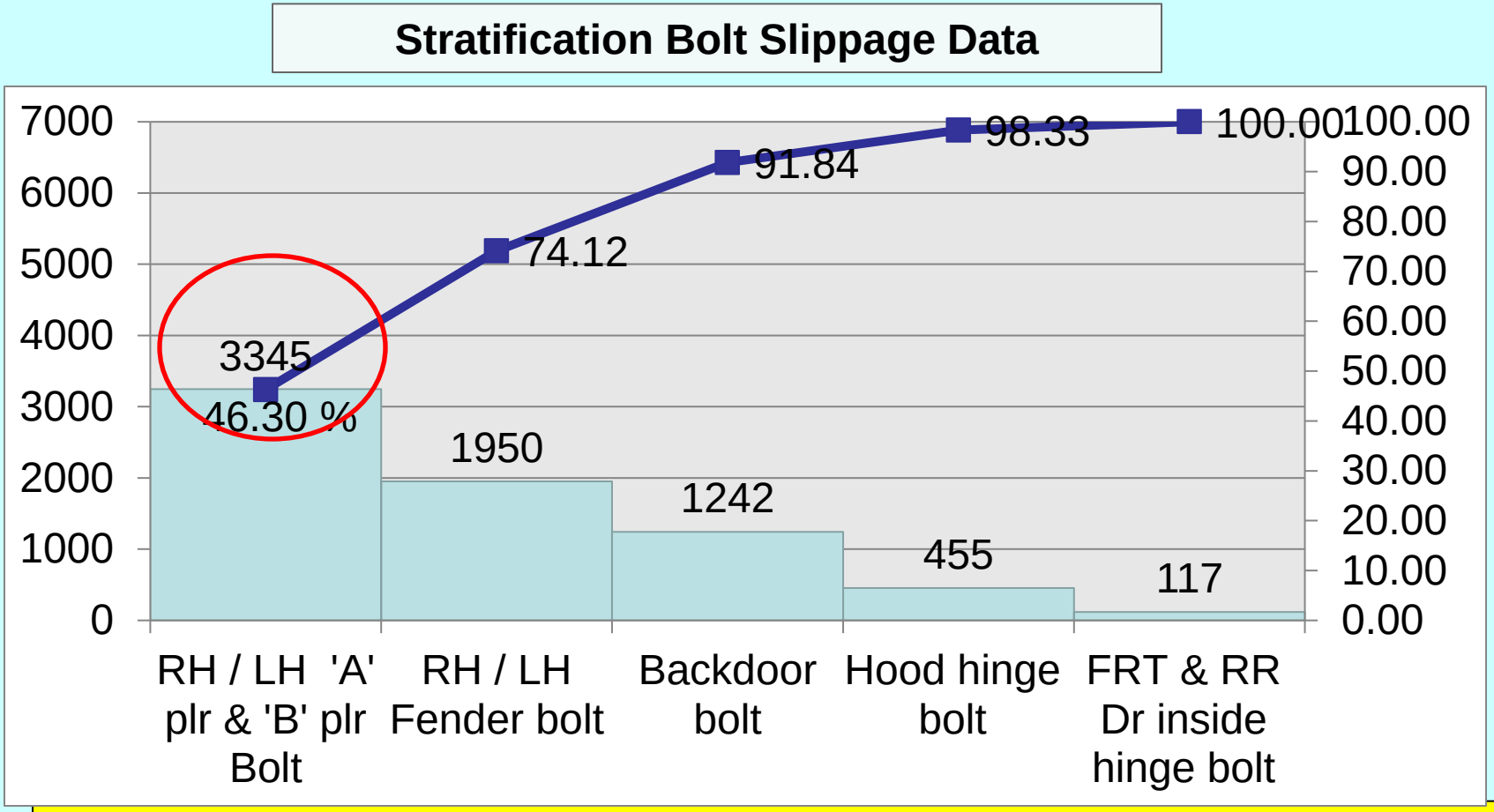
How: **Avg. @ 32 incidences per day**

2. Justify the choice (reasons)



Inference: Concern of Bolt slippage contributes 17.3 % of overall concerns & total 7009 defect incidences

2. Justify the choice (reasons)



Inference: Concern of RH / LH 'A' plr & 'B' plr Bolt slippage contributes 46.3 % of all bolt slippage concerns & total 3245 defect incidences

2.a Immediate Containment Action

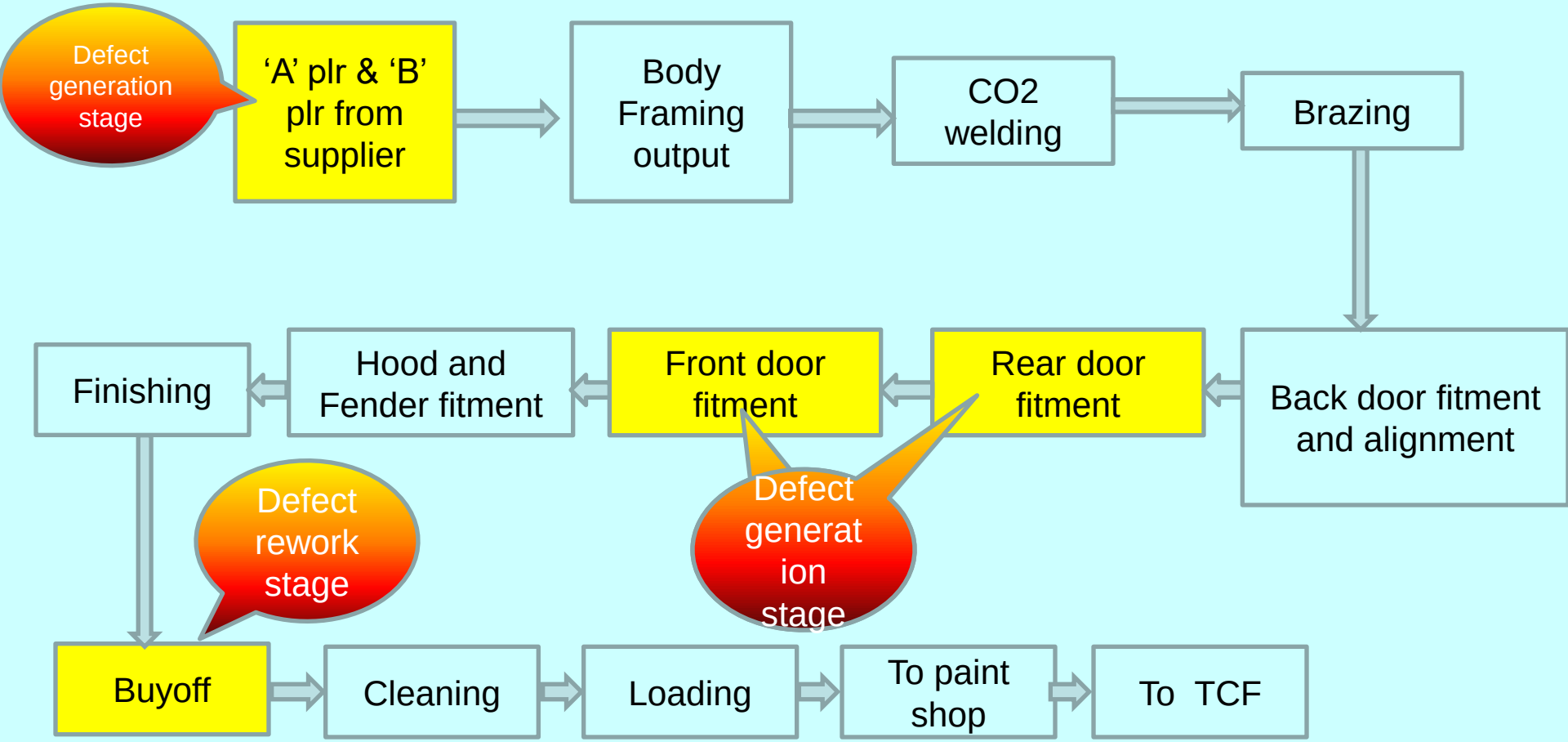
1. 100 % tapping to nut by supplier at supplier end & marking after that

2. 100 % check after closure fitment stage by check man

3. 100 % check at Buy Off & repair by Buy off operator

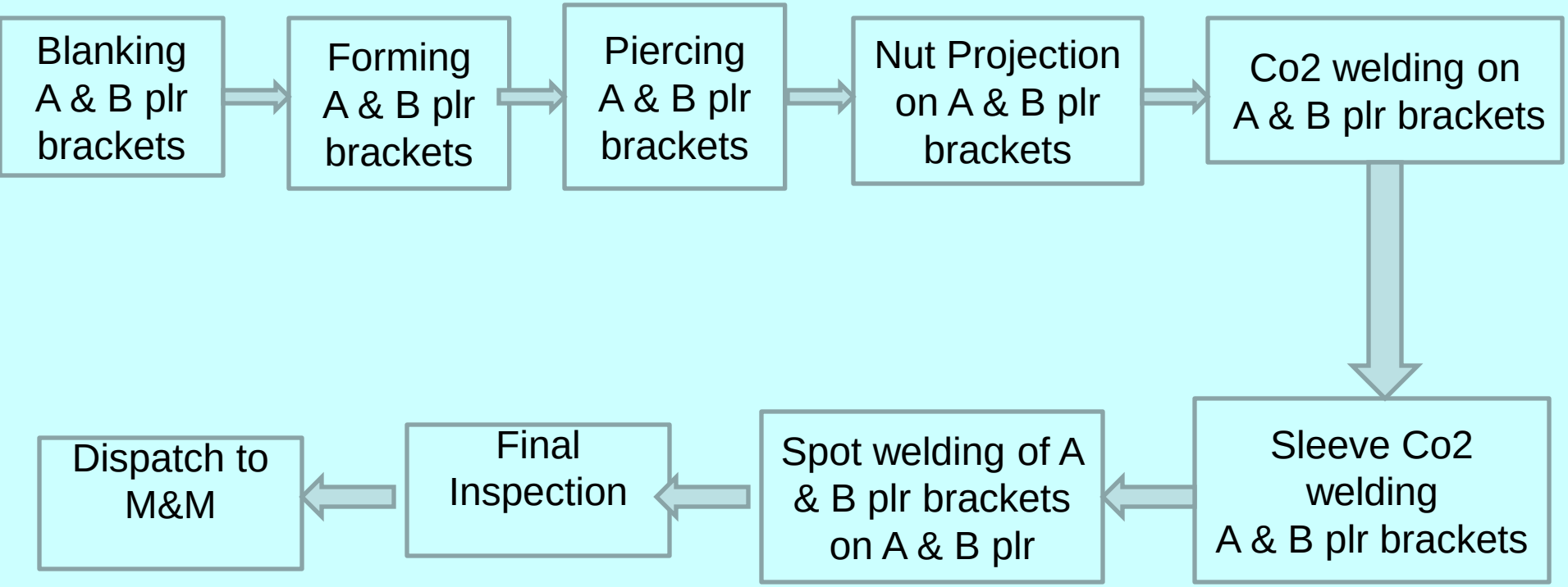
3. Understand the current situation

Observation of process:- Scorpio Bodyshop



3. Understand the current situation

Observation of process:- Harshal Pressing, Chakan



3. Understand the current situation

Observation of process:- Parts Involved

0101EG0450N	R/F ASSY PLR A LH	
0101EG0490N 0101EG0410N	R/F ASSY PLR B LH	

Step 3. Understanding the problem

D. COPQ (Cost of Poor Quality) Calculation

Description	Mar-12	Apr-12
Number of pieces rejected last month (for the part number identified for study)	– –	– –
Number of pieces scrapped last month (HEX SCREW M8X1.25X55)	730	750
Number of pieces(Vehicle) reworked last month	790	778
Part cost/piece	5.07	5.07
Rework cost/piece(Vehicle) Including (Labor Cost)	46.5	46.5
Total scrap cost (Rs.) for last month	3701.1	3802.5
Total rework cost(Rs.) for last month	36735	36177
Total Rejection cost (Rs.) for last month (Part Cost + Total rework cost)	40436.1	39979.5
Extrapolated Total rejection cost (Rs. for one year) (Average of 2Month-( ) Total-12Nos	482493.6	

4. Select Target

Present Level:	780 Incidences based on average data of last 6 months
Target:	1.156 based on average data of last 2 months 2.80 % reduction in FRC 3. Zero concern at TCF, CAI, RFI, PDI

5. Analysis

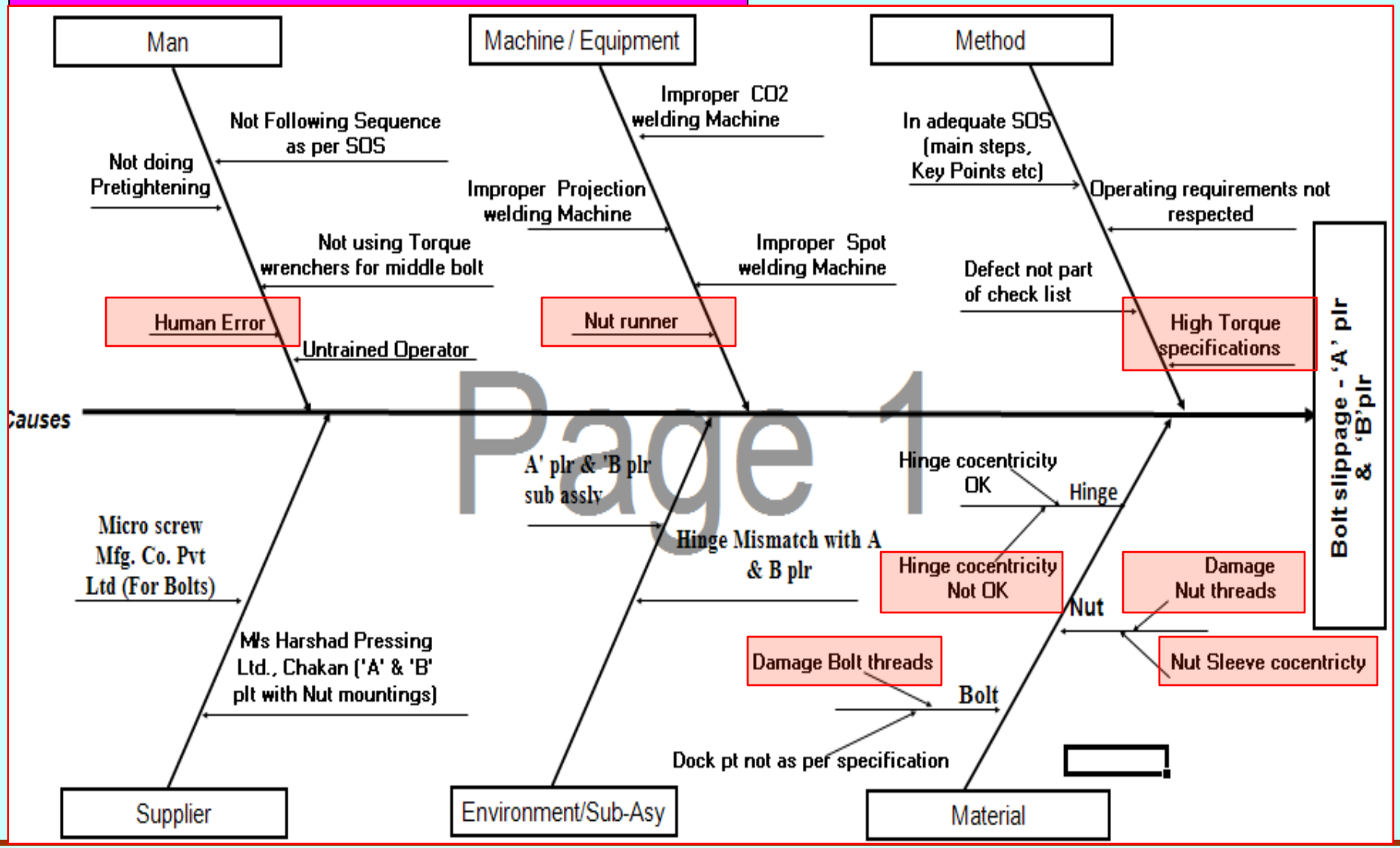
Brain Storming:

No	List of all possible causes
1	Nut thread damage
2	Nut Sleeve concentricity
3	Hinge Mismatch
4	Bolt thread damage
5	Bolt Dock pt not as per specification
6	Improper CO2 welding Machine
7	Improper Projection welding Machine
8	Improper Spot welding Machine
9	In adequate SOS (main step, Key Points etc)
10	Defect not part of check list

No	List of all possible causes
11	Operator Not Following Sequence as per SOS
12	Nut runner issue
13	High Torque specification
14	Human Error at Scorpio Body shop
15	Human Error at Supplier end
16	operator Not using Torque wrenches for middle bolt
17	Operator Not doing pretightening
18	Untrained Operator
19	A' plr & 'B plr sub assly not as per design
20	Transportation issue

5. Analysis

5.1 Identify possible causes:



Page 1

5. Analysis

List Of Probable Causes:

No	Probable Cause
1.	**Nut thread damage**
2.	**Nut Sleeve concentricity**
3.	**Hinge Mismatch**
4.	**Bolt thread damage**
5.	**Nut runner issue**
6.	**High Torque specification**
7.	**Human Error**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1.	Nut thread damage	Check 100 loose parts , assemble parts & found 67 parts Bolt not entering	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2.	Nut Sleeve concentricity	Check 100 parts & found, bolt is not entering concentrically even on the loose parts. (100%)	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
3.	Hinge Mismatch	Check mismatch of Hinge hole to Nut hole & found ; concentricity is not there for 100 % RR DR	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
4.	Bolt thread damage	Bolt thread check as per specification & found some issues with 2-3 thread after studying scrap bolts. **(91 bolts out of 100 Bolts)**	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5.	Nut runner issue	Nut runner Cp/ Cpk check & found OK	Invalid

Sr. No.	Parameter monitored	Spec.	Tool No.	Z / CC / SC	SPC TOOL USED				Monthly Cp / Cpk monitoring
	Product characteristics				Type of SPC Chart	Freq.	Measuring instrument	Cp / Cpk	
1	Front Door Lh (Closure)	25+/-5 N-m	C-369417	SC	X bar & R chart	Daily	ACTA	**Cp**	**2.43**
								Cpk	2.21
3	Rear Door Lh (Closure)	25+/-5 N-m	C-9601795	SC	X bar & R chart	Daily	ACTA	**Cp**	**2.69**
								Cpk	2.51
2	Front Door Rh (Closure)	25+/-5 N-m	SC-24	SC	X bar & R chart	Daily	ACTA	**Cp**	**2.64**
								Cpk	2.4
4	Rear Door Rh (Closure)	25+/-5 N-m	SC-23	SC	X bar & R chart	Daily	ACTA	**Cp**	**2.78**
								Cpk	2.57

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
6.	High Torque specification	Torque specification check & found OK	Invalid

Frt LH

9-Jun	10-Jun	11-Jun	12-Jun	13-Jun	14-Jun	15-Jun
25	27	25	26	25.41	25	26
26	26	24	26	26.03	25	26
26	25	24	24	25.55	26	24
25	25	26	25	25.32	24	26

Frt RH

9-Jun	10-Jun	11-Jun	12-Jun	13-Jun	14-Jun	15-Jun
25	26	25	26	24.99	26	26
24	26	25	25	25.03	26	24
26	25	25	26	25.25	25	26
26	26	24	25	25.51	26	25

RR LH

9-Jun	10-Jun	11-Jun	12-Jun	13-Jun	14-Jun	15-Jun
25	26	26	25	26.35	26	25
25	25	26	25	25.91	26	25
26	26	25	25	25.68	25	26
24	25	26	24	25.33	26	24

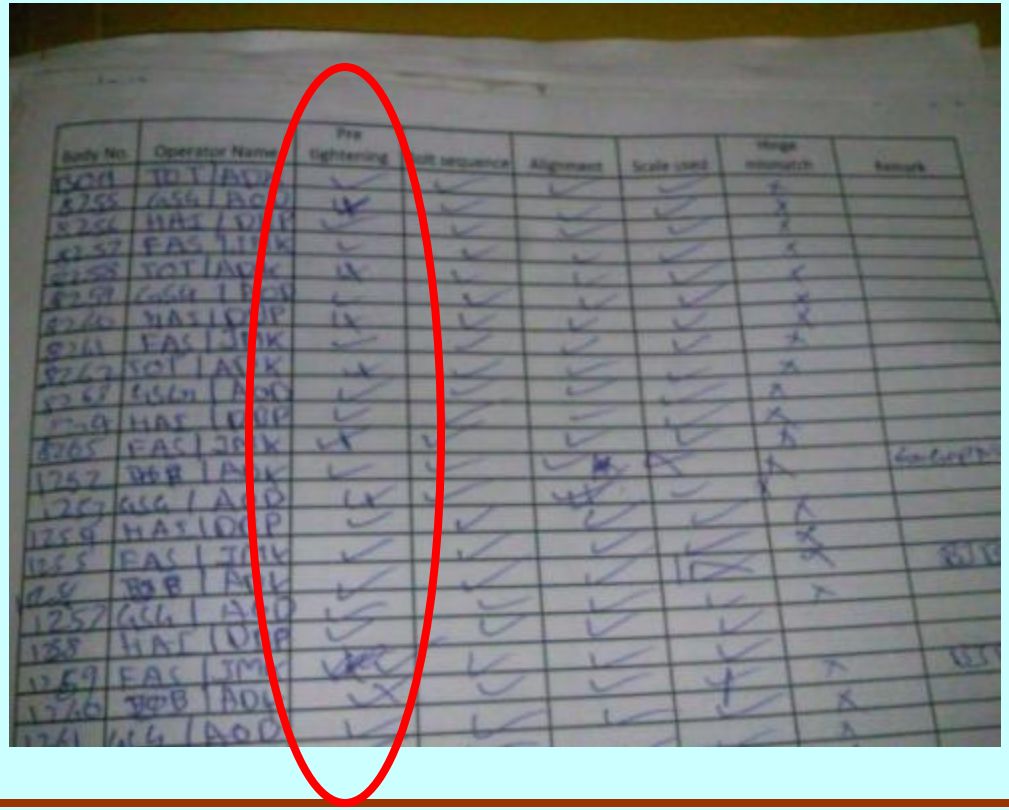
RR RH

9-Jun	10-Jun	11-Jun	12-Jun	13-Jun	14-Jun	15-Jun
26	25	26	25	25.68	26	25
25	25	26	25	26.03	26	26
25	26	25	25	25.98	25	24
24	25	26	26	25.96	26	25

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
7.	Human Error	Third party observation done & found some Operators not doing pretightening	**Valid**



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1.	Nut thread damage	Check 100 loose parts & found In most of the parts Bolt not entering	**Valid**
2.	Nut Sleeve concentricity	Check 100 parts & found, bolt is not entering concentrically even on the loose parts	**Valid**
3.	Hinge Mismatch	Check mismatch of Hinge hole to Nut hole & found ; concentricity is not there	**Valid**
4.	Bolt thread damage	Bolt thread check as per specification & found some issues with 2-3 thread	**Valid**
5.	Nut runner issue	Nut runner Cp/ Cpk check & found OK	Invalid
6.	High Torque specification	Torque specification check & found OK	Invalid
7.	Human Error	Third party observation done & found some Operators not doing pretightening	**Valid**

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	Why 1	Why 2	Why 3	Why 4
1	Nut thread damage	CO2 Weld spatter accumulation inside the nut	Direct exposure of nuts while CO2 welding	Nut thread not protected for spatter entry during CO2 Welding	RC1:-No fixture
2.	Nut -Sleeve not concentric	1) Uneven CD in-between sleeve & out of tolerance (+/- 0.4mm)	Sleeve welding fixture & Pins worn out	Old fixtures	RC2:-Fixture preventive maintenance not done & lot of tempering on fixture
		2) Nut Mounting hole deformed & Heavy Burr	1.Blank & pierce punch wear out 2. Die plate bend	RC3:-Old worn out die & die correction not possible	
		3) Nut Mounting surface not flat			

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	Why 1	Why 2	Why 3	Why 4
3.	Hinge Mismatch	Fixture pin more play	Movable pin	RC-4:-Less guide length	
4.	Bolt thread Quality issue	Initial threads of bolt worn out	Rolling dies worn out	RC-5:- Rolling dies not replaced as per defined frequency	
7.	Human Error	Operator not doing pre-tightening	Not following SOS	RC-6:- Operator taking short cut to complete the task rapidly.	

6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp	Status
1. No fixture for preventing CO2 weld spatters entrapment inside Nut threads	Fixture to be made to prevent spatter entry during CO2 Welding. (Nut Thread Protection)	01.07.12	GONSALVES S.	Closed

Before

After

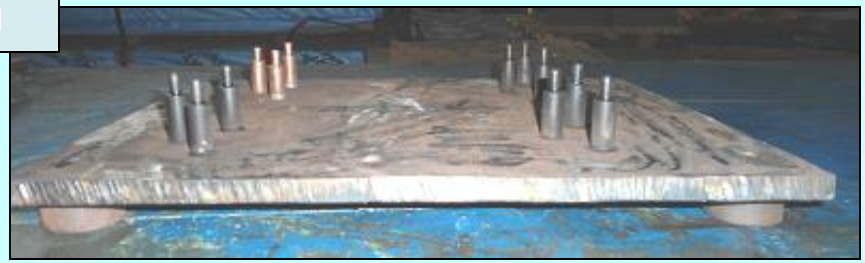
6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp	Status
1.No fixture for preventing CO2 weld spatters entrapment inside Nut threads	Fixture to be made to prevent spatter entry during CO2 Welding. (Nut Thread Protection)	01.07.12	GONSALVES S.	Closed

Before

Kaizen on Kaizen

After



6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp.	Status
2.Fixture preventive maintenance not done & lot of tempering on fixture	Complete Road map for Fixture preventive maintenance for all fixtures to me made & MIS-Q to be followed	18.07.12 For plan 01.08.12 For review	GONSALVES S.	Closed

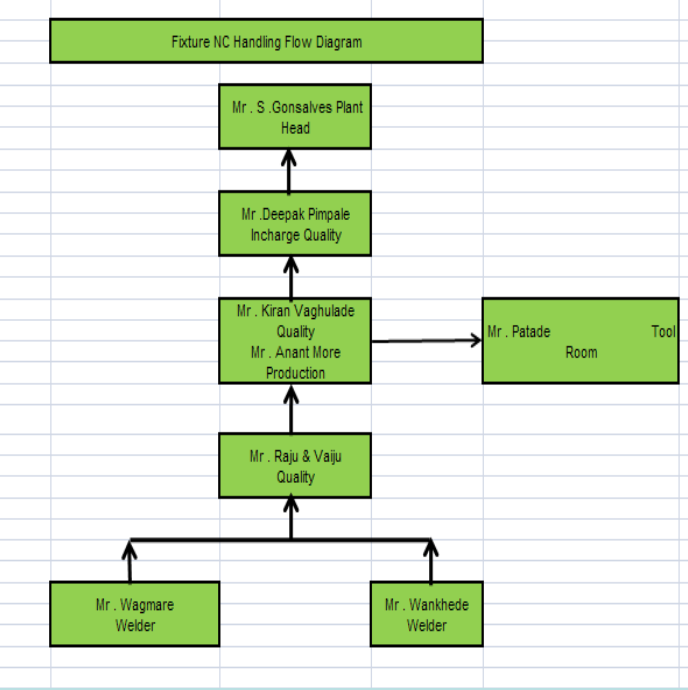
HARSHAL PRESSING PVT LTD II

VERIFICATION PLAN OF WELD FIXTURES 2012 - 2013

Sr. No.	Welding Fixture	Part Name	Verification Plan	Next Verification
1	Assy A Pillar LH/RH	Assy A Pillar LH/RH	01 - 30 Aug. 2012	01 - 30 Aug. 2013
2	Assy B Pillar LH/RH	Assy B Pillar LH/RH	01 - 30 Aug. 2012	01 - 30 Aug. 2013
3	Assy A Pillar Check arm	Assy A Pillar Check arm	1 - 30 Sep . 2012	1 - 30 Sep. 2013
4	Assy B Pillar Check arm	Assy B Pillar Check arm	1 - 30 Sep . 2012	1 - 30 Sep. 2013
5	Assy B Pillar D Loop LH	Assy B Pillar D Loop LH	1 - 30 Oct. 2012	1 - 30 OCT. 2013
6	Assy B Pillar D Loop RH	Assy B Pillar D Loop RH	01 - 30 Nov. 2012	01 - 30 Nov. 2013
7	Sleeve Tak welding Fixture	Sleeve Tak welding Fixture	01 - 30 Nov. 2012	01 - 30 Nov. 2013
8	Nut CD Insp gauge	Nut CD Insp gauge	1 - 30 DEC. 2012	1 - 30 DEC. 2013
9	Sleeve CD Insp gauge	Sleeve CD Insp gauge	1 - 30 DEC. 2012	1 - 30 DEC. 2013
10	Nut Tak Welding Fixture	Nut Tak Welding Fixture	1 - 30 DEC. 2012	1 - 30 DEC. 2013

Prepared By:- Kiran Vaghulade APPROVED BY : Deepak pimpale

WELD FIXTURES ROAD MAP 2012 - 2013



6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp.	Status
2.Fixture preventive maintenance not done & lot of tempering on fixture	Complete Road map for Fixture preventive maintenance for all fixtures to me made & MIS-Q to be followed	18.07.12 For plan 01.08.12 For review	GONSALVES S.	Closed

HARSHAL PRESSINGS PVT. LTD. II

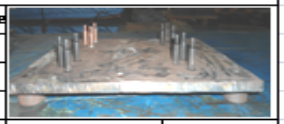
VELDING FIXTURE VERIFICATION REPORT

PART NAME: Nut Tak Welding Fixture

PART NO :

FIXTURE NO :

VARIFICATION DATE :



Sr. No	PARAMETER	SPECIFIED	TOLERANCE	OBSERVED	REMARK
1	PIN DIA	12.0	± 0.1		
2	PIN DIA	6.5	± 0.1		
3	NO OF PIN	12 Nos			
4					
5					
6					
7					

VISUAL:

Sr. No	OBSERVATION	REMARK
1	Fixture Number & Part Name	
2	Colour	
3	Damage	
4	Toggles	
5	Numbering	
6	Gap Identification	

VERIFIED BY

QA - 01/04/0-01-01-2005

HARSHAL PRESSINGS PVT. LTD. II

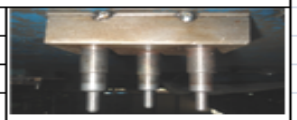
VELDING FIXTURE VERIFICATION REPORT

PART NAME: Sleeve CD Insp gauge

PART NO :

FIXTURE NO :

VARIFICATION DATE :



Sr. No	PARAMETER	SPECIFIED	TOLERANCE	OBSERVED	REMARK
1	PIN DIA	14.0	± 0.1		
2	PIN DIA	12.5	± 0.1		
3	PIN DIA	6.0	± 0.1		
4	PIN DIA	6.5	± 0.1		
5	NO OF PIN	3 Nos			
6					
7					

VISUAL:

Sr. No	OBSERVATION	REMARK
1	Fixture Number & Part Name	
2	Colour	
3	Damage	
4	Toggles	
5	Numbering	
6	Gap Identification	

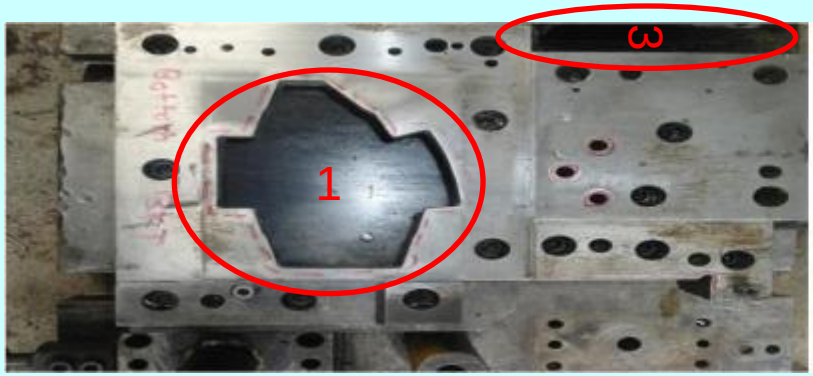
VERIFIED BY

QA - 01/04/0-01-01-2005

6. Implement the corrective measures

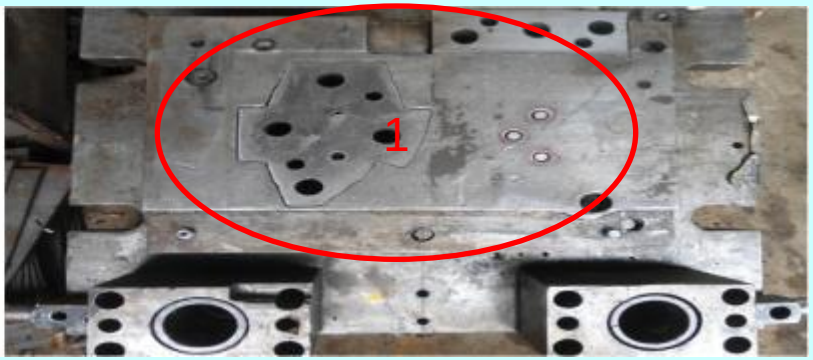
Root Cause	Action Plan	Target Date	Resp.	Status
3.OLD Worn Out Die & Die Correction not possible	Complete new die change over plan	15.07.2012	GONSALVES S.	Closed

Hinge Bottom LH /RH: 0101EG2270N / 2260N



Lower

Issue:- 1.All trim line cutting land end.
2.Boster plate bend.
3.Notch cutting land end.



Upper

Issue:- 1.Blank & pierce punch wear out.
2.Die pad Bend.

6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp.	Status
OLD Worn Out Die & Die Correction not possible	Complete new die change over plan	15.07.2012	GONSALVES S.	Closed

Hinge Bottom LH /RH: 0101EG2270N / 2260N

Stage:- Forming Tool

Lower

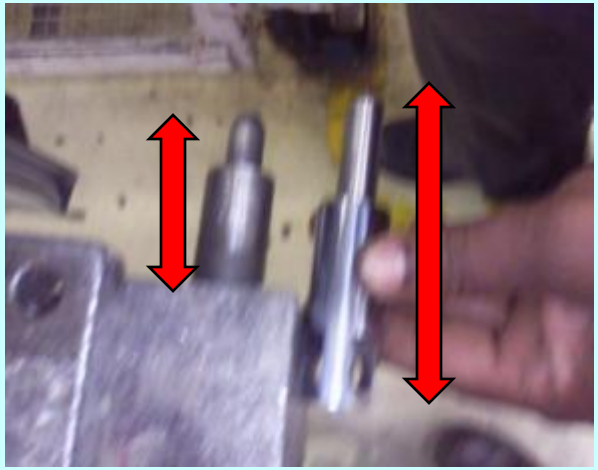
Upper

Issue:- 1.All insert welded.
2.Forming pad all portion welded.
3.Die plate bend.

Issue:- 1.All insert welded.
2.Forming pad all portion welded.
3.Die plate bend.

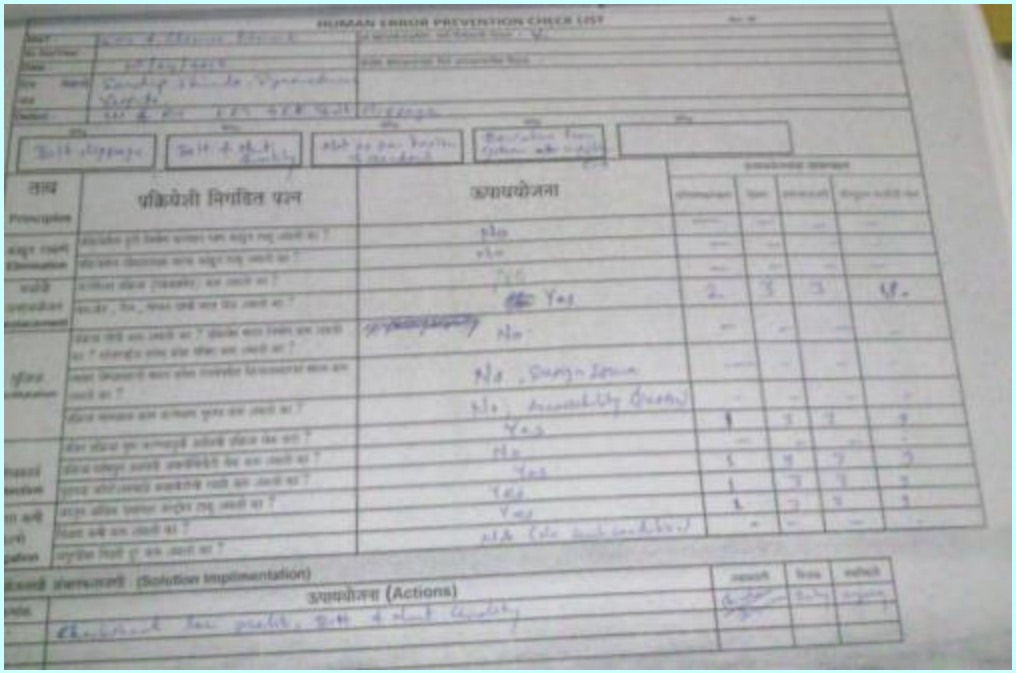
6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp.	Status
4.Hinge mismatch due to more play of movable pin having Less guide length	Movable pin to provide more guide length so that play will reduced & hinge hole concentricity with Nut hole can be achieved	06.06.2012	Hasme / Bhat VIvek	Close



6. Implement the corrective measures

Root Cause	Action Plan	Target Date	Resp.	Status
5.Initial thread of bolt worn out due to Rolling dies not replaced as per defined frequency	Rolling dies to be replaced	25.06.2012	Ketan Wagh & Micro screw Mfg. Co. Pvt Ltd.	Close
6.Operator taking short cut to complete the task rapidly.(Human Error)	100 % cross audit for 1 month & then to cross verify through 2A-2B audit, Supervisory check sheet	15.06.2012 & ongoing	Shinde S S & Singh Jagveer	Close & ongoing



Human error prevention check sheet

6. Implement the corrective measures

OTHER ACTIONS :-

Quality Issue:- Projection nut welding pin is of Bakelite & worn out.

Before

Bakelite pin used for nut projection.

After

SS Pin to be used in-place of Bakelite Pin.

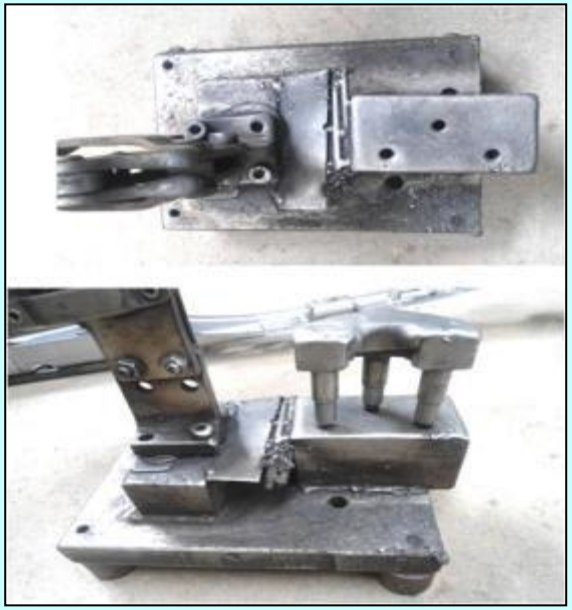
Cu Pin to be used in-place of SS Pin.

6. Implement the corrective measures

OTHER ACTIONS :-

Quality Issue:- Sleeve welding fixture worn out.

Before



Sleeve locating pins & block play observed.

After

Sleeve locating pins & block replaced new.

6. Implement the corrective measures

OTHER ACTIONS :-

Quality Issue:- No Inspection method for nut offset w.r.t. nut.

New gauge made for Nut offset w.r.t. Nut inspection.

6. Implement the corrective measures

OTHER ACTIONS :-

Quality Issue:- Welding fixture pin loose, Pin not Concentric.

Before

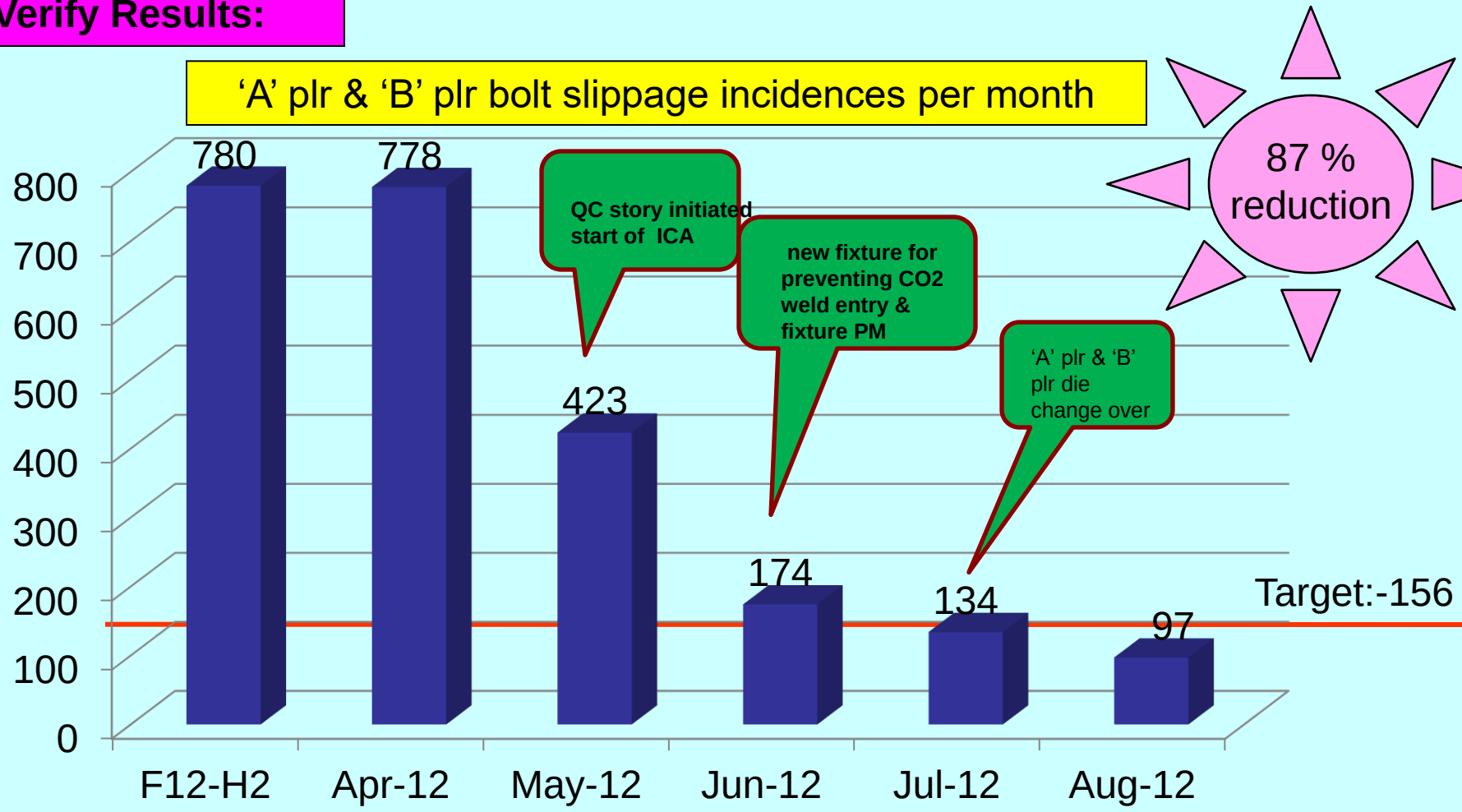
1.B pillar spotting welding fixture pins loose observed.
2.Pins welded on the plate.

After

1.B pillar spotting welding fixture with pins made new.
2.Pins bolting on the plate.

7. Check & confirm the results & effects

7.1 Verify Results:



7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

1.All decided actions implemented as on 30.07.2012

2. Effectiveness of actions has been audited on 02.08.2012 by SQA team

3. Frequent Supplier audit by CFT.

Check & confirm the results & effects (Cont……)

7.3 Tangible benefits:

1.'A' plr & 'B' plr M8 Hinge Bolt Slip incidences

reduced from 780 per month to 97 per Month

2. Zero incidences at TCF,CAI,RFI,PDI from last 3 months.

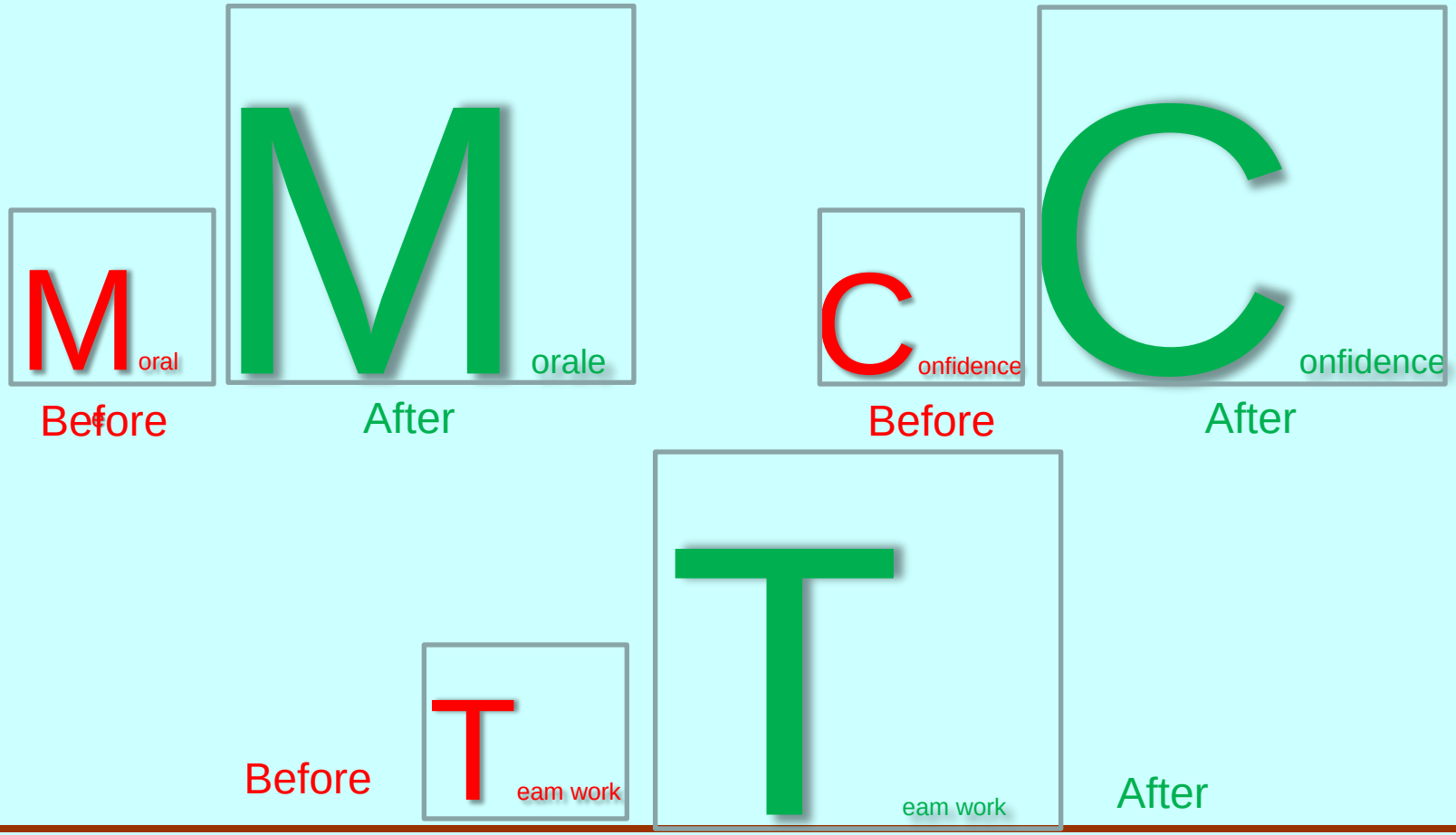
3. Cost saving benefits :
Saving in scrap cost of bolt= Rs. **5.07** per incidence i.e. Rs.**45021** per year.

Saving in Rework cost = **Rs. 437472/ Year**

Total Cost saved = Rs.482493.6/Year

7.3 Intangible benefits:

Team Morale , Team Work & confidence level to solve chronic concern is improved substantially.



8. Standardisation

8.1 Updation of documents:

1.Control Plan

Plant : HARSHAL PRESSINGS PVT. LTD.	Model : Scorpio	Mahindra & Mahindra Control Plan			Cross Functional Team
Part No/s.:0101EG2490N	Part Name/ Process: REINF FRT HINGE BOTTOM				Mr. S.Gonsalvis,Mr. Deepak Pimpale,Mr.Sachin Bhosale, Mr.Kiran waghulde,Mr.vaijayanti.jagtap
Latest Change Letter:	Doc No.: CP/m&m/0101EG2490N	Prototype : ☐	Prelaunch: ☐	Production: ☐	Critical Char: <CC>
Aggregate No.:	Aggregate Part Descrptn :	Module Name :	Reviewed by :	Rev. No : B	Significant Char : <SC>
PU/ Shop :	Next Aggregate No.:	Cell Name :	Approved by :	Rev. Date :	CP Date(Orig) : 10.07.2012

Opn No.	Operation Description	Machine, Device, Jig, Tools Description	Characteristics No.	Characteristics Product	Characteristics Process	Sp. Chara. Class	Method Product/ Process Specification.	Method Evaluation Measurement Tech.	Sample Size	Sample Freq.	Control Method Record (Y/N)	Control Method Method	Control Method Resp.	Reaction Plan
10	Raw Material Inspection	Manually	1	Mat. Thickness		--	1.6 ± 0.11	Micrometer (0-25mm)	Micrometer (L.C.- 0.01 mm)	/ 1 Hr.	Y	QA I/C	QA I/C	Reject the lot & Inform to Purchase
			2	Mat . Grade		--	D 513	External Chem.lab	Supplier Test Certificate/ Lab Test Report	Per Lot	Y			
			3	Surface Condition		--	Free from rusty,Damage,Pitted,Black shade etc	Visual	100%	Per Lot	Y			
20	SHEARING	Shearing m/c	1	Length	Stopper & Blade sharpness.		1250 ± 1.0	Measuring tape	5Nos	/ 1 Hr.	Y	1st Piece & Inprocess Inspection	Supervisor	Reject the lot & Inform to Supervisor
			2	Width			195 ± 1.0	Measuring tape						
			3	Condition			Free from burr	Visual	100%	Per Lot				
30	BLANK	Use Blank tool	1	Dimn			170.0±0.5	Height Gauge(L.C. 0.02 mm)	5 Nos	/1Hr.	Y	1st Piece & Inprocess Inspection	Supervisor	Reject the lot & Inform to Supervisor
			2	Dimn		--	169.0±0.5							
			3	Burr,Dent& damage			Free From Burr, ,Dent& damage	Visual						
40	FORMING	Use Forming tool	1	Dimn			19.80±0.2	Vernier Caliper (L.C.- DV 0.01 mm)	5 Nos	/1Hr.	Y	1st Piece & Inprocess Inspection	Supervisor	Reject the lot & Inform to Supervisor
			2	Dimn		--	19.50±0.2	Vernier Caliper (L.C.- DV 0.01 mm)						
			3	Dimn			18.0±0.2	Vernier Caliper (L.C.- DV 0.01 mm)						
			4	Profile			To Qualify with checking gauge	Visual						
			5	Scoring, scratches			Free From Scoring, scratches	Visual						
50	PIERCE	Use Trim & Pierce tool	1	2 C.D.			20.0±0.2	Vernier Caliper (L.C.- DV 0.01 mm)	5 Nos	/1Hr.	Y	1st Piece & Inprocess Inspection	Supervisor	Reject the lot & Inform to Supervisor
			2	4 C.D.		--	27.0±0.2	Vernier Caliper (L.C.- DV 0.01 mm)						
			3	3 Holes Dia			10.5+0.3	Vernier Caliper (L.C.- DV 0.01 mm)						
			4	Burr,Dent& damage			Free From Burr, ,Dent& damage	Visual						

ev No	Rev Date	Revision Summary	Reason for Revision

PREPARED BY :	APPROVED BY :	Date:-	Control status

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8. Standardisation

8.1 Updation of documents:

2. FMEA

Plant : HARSHAL PRESSINGS PVT. LTD.		Model : Scorpio			Mahindra & Mahindra										Cross Functional Team				
Part No/s.:0101EG2130N / 2120N		Part Name/ Process: REINF FRT HINGE TOP LH / RH													Mr. S.Gonsalvis,Mr. Deepak Pimpale,Mr.Sachin Bhosale, Mr.Kiran waghulde,Ms.vaijayanti.jagtap.				
Latest Change Letter:		Optn No.:			Process Potenial Failure Mode & Effect Analysis.														
PU/ Shop :		Optn Desptn:			Reviewed by :			Rev. No : B				Doc No.: Pfmea/m&m/0101EG2130N							
Module/ Cell :		M/C Name & No.:			Approved by :			Rev. Date :				FMEA Date (Orig) : 01.07.2012							
Process Function / Requirements	Potential Failure Mode	Potential Effect (s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Occur	Current Process Controls Prevention	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
RAW MATERIAL INSP.	1) Wrong material thk 2)Wrong material grade 3) Waviness	1)Component doesn't meet Drg.requirement	6	NA	1)Material mixed up 2)Wrong P.O. 3)old material 4) loose material packing	2	1)Send correct po to supplier as per RMQP 2)Std. Packing SOP 3)Use fifo systeam	1) Incoming inspection report 2)Verification of material TC	4	48							
SHEARING	1)Under size cutting	1) Comp. Rejected	5	NA	1) Stopper setting is wrong	4	1)Use work instruction	First piece / Last Piece / In process Inspection/Visual Inspection	4	80							
	2)over size cutting	2) Fitment problem in Subsequent Operation	5		2) Stopper setting is wrong	4			4	80							
	3)Teper cutting	3) Comp. Rejected	5		3) Stopper setting is wrong	4			4	80							
	4)Burr	4)Inconvenence to the handle 5)Gap between mating parts	5		4)M/C setting not Proper / Shearing Blead get blant 5)Excess clearance between die	2	2)CLEARENCE CHECKING RESHARPENING OF BLADE AS PER TOOL MAINTAINANCE PLANE		4	40							
BLANK	1. Burr	1.Poor appearance	4	NA	1) Excess clearance between die&punch	2	During Preventive Maintainance checking of Clearance of die & punch	First piece / Last Piece / In process Inspection FP/IP/LP-QA/13/00	6	48							
		2) Unsafe To operator	6		2) Blunt cutting edges.				4	72							
	2. Damage	1.Surface mismatch	6		1.Mislocation of part by operator	2	Location pin provided.		5	60							
	3.Dent		6		2.Foregin particle on die Surface	3	2)Periodic Resharpening of tool		5	90							
	4. Off Cut	1)Rejection at next Operation	7		1) Alignment of tool not proper / wear of stopper	2	1)Preventive maintenance of tool		5	70							
FORM	1) Improper Comp Restrike	1) Component reject.	5	NA	1) Stopper setting is wrong	3	1) Set stoper properly	First piece / Last Piece / In process Inspection/Visual Inspection	4	60							
	2) Scratches & Scoring Marks	2) Aesthetically not ok	3		2) Improper clearance 3)Foregen particles in the die	4	2)Preventive maintenance of tool 3)Polishing of tool as per P. M. checklist		4	48							
	3)Wrinkles	Component rejected	7		1) Improper tool setting Pressure & cushion pressure stroke less than REQUIREMENT) 2) Improper Holding 3) Excess Oil Application	2	1) Tool set as per setting chart (Cushion Pressure & Cushion should be as per given should be as per given setting chart 2) varify as per setup approval record 3)Oil pattern to be used as per oil pattern layout given in setting chart.		4	56							
	4))Crack & Thinning	3) Component rejected	6		3) Stroke length not proper & Unequal cushion pressure, Die polishing not ok.	2	3) Set die & punch properly. Preventive maintenance of tool Cleaning & polishing of tool		4	48							
PIERCE	1) Dimension variation	1)Fitment problem at customer end.	6		1) Alignment of tool not proper / wear of stopper	2	1)Preventive Maintenance of tool	First piece / Last Piece / In process Inspection	5	60							
	2) Hole under size	3)Fitment problem in Subsequent Operation	6	NA	3)Wear of punches	3	3)Preventive maintenance of tool		5	90							
	3)Hole missing	4)Fitment problem in Subsequent Operation	6		4)Punch broken	2	4)Locating pin in the next operation		5	60							
	4) Wrong Hole Position / Piercing Shift.	5)Fitment problem in Subsequent Operation	5		5)No proper stopper provided in the tool	3	5)Proper stopper /Proper Gauging provided in the tool (Fool- Proofing)		5	75							
	5) Burr	1.Poor appearance	4		1) Excess clearance between die&punch	3	During Preventive Maintainance checking of Clearance of die & punch		5	60							
		2)UnsafeTo operator	7	NA	2) Blunt cutting edges.	3			4	84							
Projection welding M/C	1)Nut Shifting	Fitment problem at coustomer end	6	NA	1)Inproper loction in projection	3	1)Provided location for nut Projection	First piece / Last Piece / In process Inspection/Visual Inspection 100%	4	72							
	2)Welding spatter		6		2)Welding parameter not set	2	2.a)Monitioring of weld seting parameters. 2.b)Carbon test after every two hours. 3.c)Torque test teken 05		5	60							
	3)Nut remove in assy.		6			2			5	60							
	4)Nut missing		6		3)Opretor misstaek	3	Ref. sample kept on work station.		4	72							
(CO2 WELDING)	1) Weld spatter / Blow holes /Material burnt during welding	1) Component rejectedvechicle	6	NA	1)Improper weld setting parameters , Unclean material surface as like oil dust etc	3	1) Work instruction Displayed Near work station & Monitoring	First piece / Last Piece / In process Inspection/Visual Inspection 100 % visual inspection by operator	4	72							
	2) Less welding	2)Less Strength of weld joint of parts.	6		2)Operator Mistake Dressing	3	2)Ref.Sample has kept near the station		5	90							
Rev No.:	Rev Date	Revision Summary						Reason for Revision									
Note :- FMEA will be revised and appropriate action will be taken, when RPN value will be more than 100.																	
PREPARED BY :				APPROVED BY :								Date.-		Control status			

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8. Standardisation

8.1 Updation of documents:

1. Mahindra Inspection Standard – Quality for ‘A’ plr LH
2. Mahindra Inspection Standard – Quality for ‘A’ plr RH

Mahindra Inspection Standard - Quality

MIS-Q	Supplier Name				Supplier Code		PAGE
	HARSHAL PRESSING PVT. LTD.PUNE				DH006		1/1
Model	Scorpio All	PART NAME	REINF ASSY PILLAR A -LH		PART NUMBER	0101EG0440N	CL-H
S.NO.	INSPECTION ITEM	CRITERIA	INSPECTION METHOD	INSPECTION TOOL	SAMPLING PLAN	INSPECTION STAGE	REMARKS
A	FITMENT						
1	A-PILLAR & CHECKARM	10.5 ±0.5 mm	CHECK	INSPECTION GAUGE	100%	FINAL	
2	CHECKARM INNER WIDTH	7.0 ±0.2 mm	CHECK	INSPECTION GAUGE	100%	IN PROCESS	
3	CHECKARM HOLE DIA	6.0 +0.2/-0.0 mm	CHECK	INSPECTION GAUGE	100%	IN PROCESS	
4	NUT THREAD	M8*1.25	CHECK	BY BOLT TRG	100% 5 in 100Nos	INPROCESS FINAL	BOLT CHANGE FREQ 500 NOS.
5	PROFILE FITMENT	3.0±0.5 mm	CHECK	INSPECTION GAUGE	5 SET / 3 MONTH	IN PROCESS	
6	SLEEVE / NUT CONCENTRICITY	0.5mm	CHECK	THREAD PIN STEP GAUGE	100%	FINAL	
B	FINISH						
1	BURR	FREE FROM BURR	CHECK	VISUAL	100%	FINAL	
2	Rust	Free From Rust	CHECK	VISUAL	100%	FINAL	
C	FUNCTION						
1	NO. OF SPOT	12.00	COUNTING & MARKING	VISUAL	100%	FINAL	
2	Spot Nugget Dia	Nugget Dia 6 mm	MEASURE	V.C	SETUP	IN PROCESS	
3	TIGHTEN TORQUE	40-45 Nm	TEST	TORQUE WRENCH	1 /100	IN PROCESS	
4	SPOT WELD INTERGRITY	NO SPOT FAILURE	TEST	TEAR DOWN TEST	1st week of every month	FINAL	

All variable parameters should be recorded
All functional parameters should have traceability to process control records.
Process capability report should be submitted at monthly frequency.
Endurance test reports to be submitted as per frequency

Origin Date : Rev No : Rev Date :

For M&M use only :

MIS-Q Sr No : Prepared By : Approved By :

PQA/SQA/F/23 Rev : 0.00 Date : 07-Dec-10

Page 1

8. Standardisation

8.1 Updation of documents:

3. Mahindra Inspection Standard – Quality for ‘B’ plr LH
4. Mahindra Inspection Standard – Quality for ‘B’ plr RH

Mahindra Inspection Standard - Quality

MIS-Q	Supplier Name				Supplier Code		PAGE
	HARSHAL PRESSING PVT. LTD,PUNE				DH006		1/1
Model	Scorpio All	PART NAME	REINF ASSY PILLAR A -LH		PART NUMBER	0101EG0440N	CL-H
S.NO.	INSPECTION ITEM	CRITERIA	INSPECTION METHOD	INSPECTION TOOL	SAMPLING PLAN	INSPECTION STAGE	REMARKS
A	FITMENT						
1	A-PILLAR & CHECKARM	10.5 ±0.5 mm	CHECK	INSPECTION GAUGE	100%	FINAL	
2	CHECKARM INNER WIDTH	7.0 ±0.2 mm	CHECK	INSPECTION GAUGE	100%	IN PROCESS	
3	CHECKARM HOLE DIA	6.0 +0.2/-0.0 mm	CHECK	INSPECTION GAUGE	100%	IN PROCESS	
4	NUT THREAD	M8*1.25	CHECK	BY BOLT TDQ	100% 5 in 100Nos	IN PROCESS FINAL	BOLT CHANGE FREQ 100 NOS.
5	PROFILE FITMENT	3.0±0.5 mm	CHECK	INSPECTION GAUGE	5 SET / 3 MONTH	IN PROCESS	
6	SLEEVE / NUT CONCENTRICITY	0.5mm	CHECK	THREAD PIN STEP GAUGE	100%	FINAL	
B	FINISH						
1	BURR	FREE FROM BURR	CHECK	VISUAL	100%	FINAL	
2	Rust	Free From Rust	CHECK	VISUAL	100%	FINAL	
C	FUNCTION						
1	NO. OF SPOT	12.00	COUNTING & MARKING	VISUAL	100%	FINAL	
2	Spot Nugget Dia	Nugget Dia 6 mm	MEASURE	V.C	SETUP	IN PROCESS	
3	TIGHTEN TORQUE	40-45 Nm	TEST	TORQUE WRENCH	1 / 100	IN PROCESS	
4	SPOT WELD INTERGRITY	NO SPOT FAILURE	TEST	TEAR DOWN TEST	1st week of every month	FINAL	

All variable parameters should be recorded
All functional parameters should have traceability to process control records.
Process capability report should be submitted at monthly frequency.
Endurance test reports to be submitted as per frequency

Origin Date : Rev No : Rev Date :

For M&M use only :

MIS-Q Sr No : Prepared By : Approved By :

PQA/SQA/F/23 Rev : 0.00 Date : 07-Dec-10

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8. Standardisation

8.1 Updation of documents:

5. Verification plan of fixtures by Supplier

HARSHAL PRESSING PVT LTD II						
VERIFICATION PLAN OF WELD FIXTURES 2012 - 2013						
Sr. No.	Welding Fixture	Part Name	Part No	Verification Plan	verified on	Next Verification
1	Assy A Pillar LH/RH	Assy A Pillar LH/RH		01 - 30 Aug. 2012		01 - 30 Aug. 2013
2	Assy B Pillar LH/RH	Assy B Pillar LH/RH		01 - 30 Aug. 2012		01 - 30 Aug. 2013
3	Assy A Pillar Check arm	Assy A Pillar Check arm		1 - 30 Sep . 2012		1 - 30 Sep. 2013
4	Assy B Pillar Check arm	Assy B Pillar Check arm		1 - 30 Sep . 2012		1 - 30 Sep. 2013
5	Assy B Pillar D Loop LH	Assy B Pillar D Loop LH		1 - 30 Oct. 2012		1 - 30 OCT. 2013
6	Assy B Pillar D Loop RH	Assy B Pillar D Loop RH		01 - 30 Nov. 2012		01 - 30 Nov. 2013
7	Sleeve Tak welding Fixture	Sleeve Tak welding Fixture		01 - 30 Nov. 2012		01 - 30 Nov. 2013
8	Nut CD Insp gauge	Nut CD Insp gauge		1 - 30 DEC. 2012		1 - 30 DEC. 2013
9	Sleeve CD Insp gauge	Sleeve CD Insp gauge		1 - 30 DEC. 2012		1 - 30 DEC. 2013
10	Nut Tak Welding Fixture	Nut Tak Welding Fixture		1 - 30 DEC. 2012		1 - 30 DEC. 2013
Prepared By:- Kiran Vaghulade			APPROVED BY : Deepak pimpale			

8. Standardisation

8.1 Updation of documents:

6. Welding fixture verification report

HARSHAL PRESSINGS PVT. LTD. II

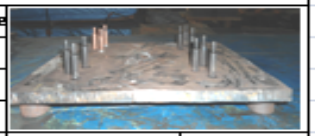
VELDING FIXTURE VERIFICATION REPORT

PART NAME: Nut Tak Welding Fixture

PART NO :

FIXTURE NO :

VARIFICATION DATE :



Sr. No	PARAMETE	SPECIFIED	OLERANCI	OBSERVED	REMARK
1	PIN DIA	12.0	± 0.1		
2	PIN DIA	6.5	± 0.1		
3	NO OF PIN	12 Nos			
4					
5					
6					
7					

VISUAL:

Sr. No	OBSERVATION	REMARK
1	Fixture Number & Part Name	
2	Colour	
3	Damage	
4	Toggles	
5	Numbering	
6	Gap Identification	

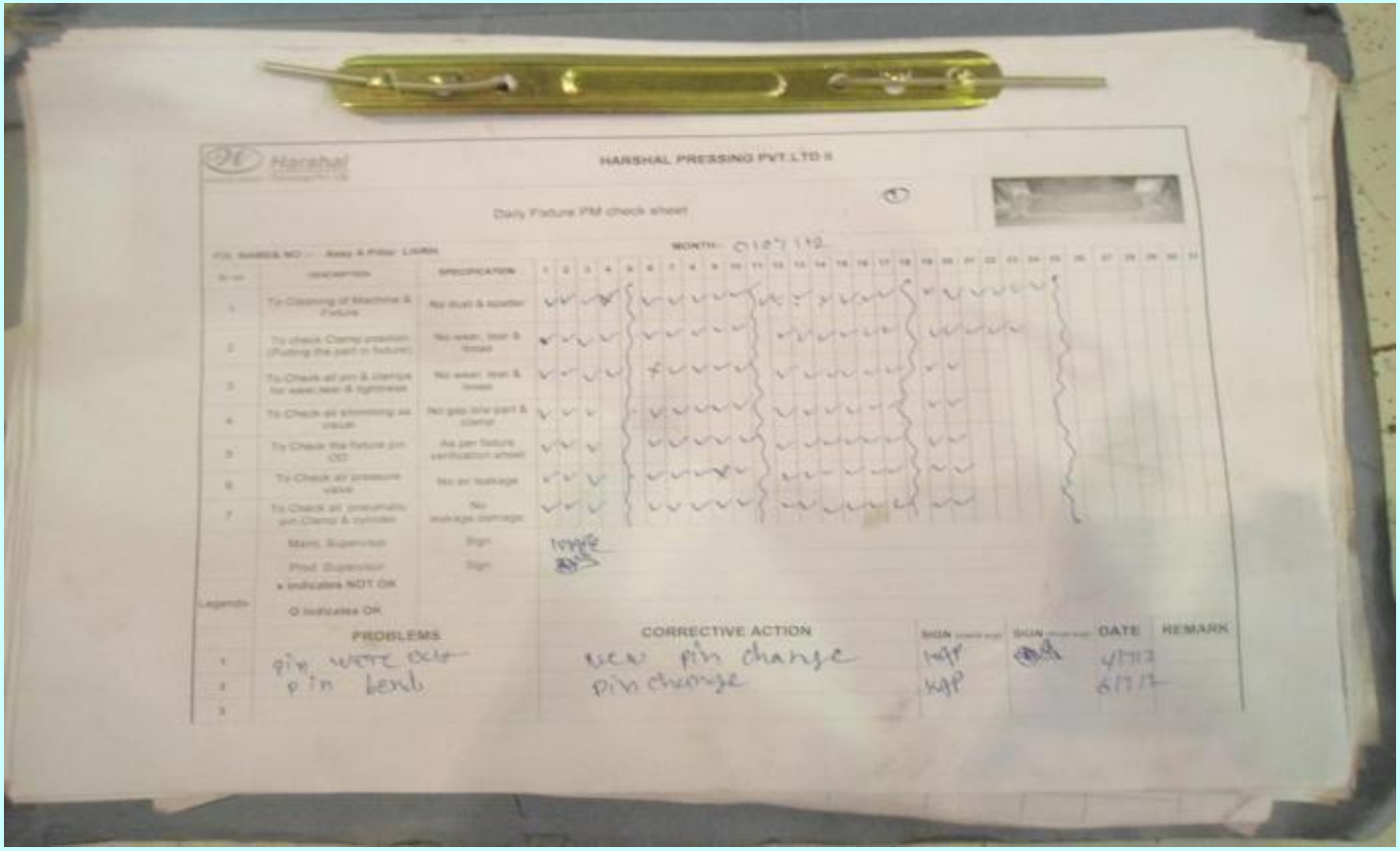
VERIFIED BY

QA - 01/04/0-01-01-2005

8. Standardisation

8.1 Updation of documents:

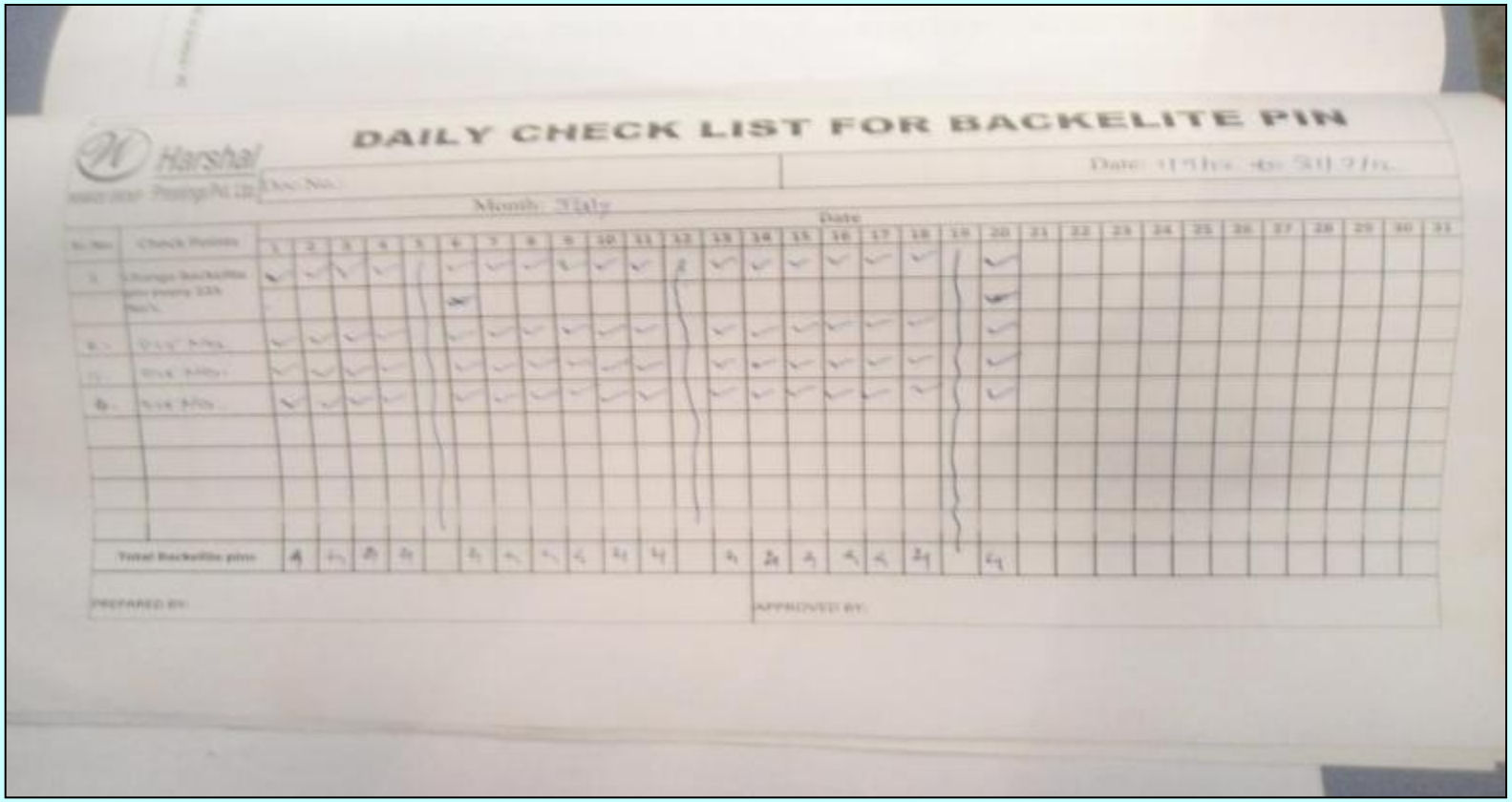
7. Daily Fixture PM



8. Standardisation

8.1 Updation of documents:

8. Daily Checklist for pins



8. Standardisation

8.2 Training :-

1. OJT for closure fitment stage operators.

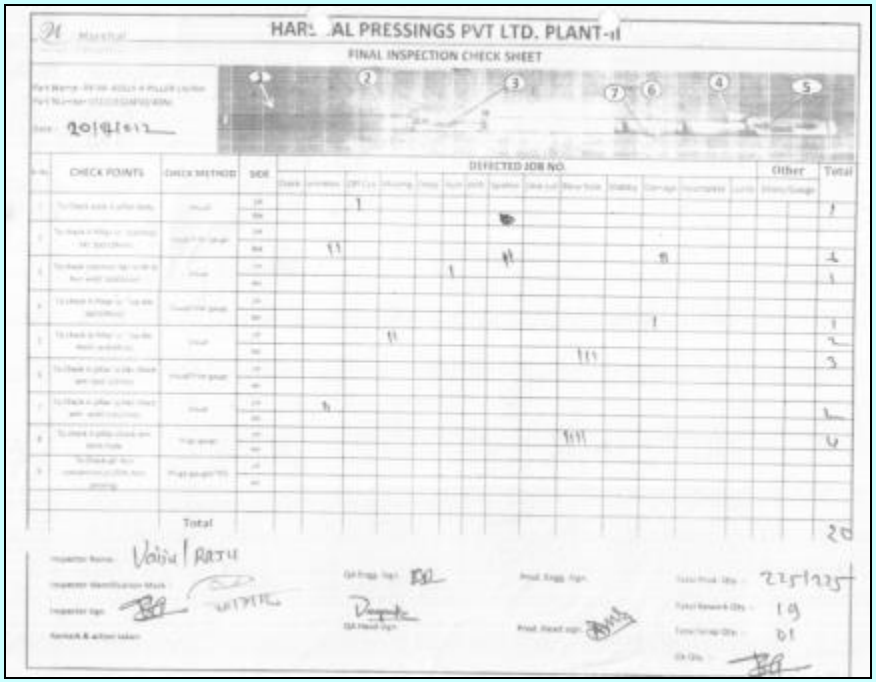
2.SOS training

3. ILU training

8. Standardisation

8.3 Additional Audit / Check

1. Supplier Audit
2. Supervisory check sheet
3. Final Inspection at Supplier End



9. Summarise & Plan future actions

9.1 Conclusion:

MYB QC story approach and team work by doing GEMBA study can lead to elimination of such chronic concerns. CO2 weld spatter entry inside the nut was the major concern which is eliminated by providing special fixture.

Concern is completely eliminated at Customer end. i.e. TCF,CAI,PDI etc.

Zero incidences at TCF,CAI,RFI,PDI from last 3 months.

QC story methodology is Highly recommended for resolving chronic concerns.

As against the Target of 80% reduction ,team have achieved 87 % reduction in incidences. Thus Target Achieved .

9. Summarise & Plan future actions

9.2 Horizontal Deployment:

1. Hood hinge M8 bolt slippage (DCR No – 21288 Dated 13.06.2012)
2. Backdoor M10 bolt slippage

   Action- Developed new back door assembly fixture to eliminate back door hinge to D plr nut concentricity issue

9. Summarise & Plan future actions

9.3 Lessons Learnt:

Things Gone Wrong :

1.Initially dependency of resolving the concern on SQA only; thus no CFT approach resulting into unsustainable results.

2. Budgetary provision on reactive mode resulting into loss of huge time span.

Things Gone Right :

1.Always challenge given inadequate process leading to innovative ideas thoughtfully implemented

2.Human Error Prevention

3.Resolution of Chronic Concern with the QC story methodology

4. CFT approach with GEMBA culture is very helpful in resolving chronic concern.

By Co creating ,Accepting no limits and Alternative thinking ,we enable our stake holders and us to

…RISE

Thank You